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EXPERIMENT TITTLE: MOMENT OF INERTIA OF A ROTATING WHEEL

Objective :

1. Determine the moment of inertia of a system consisting of a bicycle wheel

EXPERIMENT SUMMARY

INTRODUCTION

Basically, the moment of inertia of objects with regular shapes such as solid cylinders, solid spheres, and rods can be easily determined. However, it becomes quite difficult for objects with irregular shapes because the moment of inertia of a rotating wheel can be determined dynamically. In this case, the moment of inertia is obtained by observing the acceleration of an object set in motion by a load dropped from a certain height. The moment of inertia of an irregularly shaped object can be determined using the principles of mechanics, specifically Newton's Second Law, which governs translational and rotational motion. This law states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass [1]. In classical mechanics, the motion of an object is classified into two main types of translational motion and rotational motion. Translational motion occurs when all systems within an object move at the same velocity and in the same direction along a linear path. In contrast, rotational motion occurs when all systems within the object move in a circular path around a fixed axis of rotation or pivot [2]. In many real world mechanical systems, an object often undergoes both translational and rotational motion simultaneously such as a rolling wheel, where the dynamic analysis requires consideration of the center of mass and rotation relative to that center of mass.

To understand the transition from kinematics to dynamics in circular motion, one must understand the relationship between angular acceleration and tangential acceleration. When an object rotates, its angular velocity can change over time, resulting in angular acceleration (α). Every point on the object at a distance (r) from the axis of rotation will have a tangential linear acceleration (a_T) that is tangent to the object's circular path [3]. Tangential acceleration represents the change in an object's linear velocity over time, while angular acceleration represents the change in the rotational speed of the entire structure of an object within a system simultaneously.

In the discussion of the causes of rotational motion, the concept of force in translational motion is replaced by the concept of torque (moment of force). Torque (τ) is defined as the cross product of the position vector of the point of application of the force relative to the axis of rotation (r) and the applied force vector (F), such that $\vec{\tau} = \vec{r} \times \vec{F}$ [4]. Translational motion of a load is expressed in Equation 1 as mass (kg) multiplied by acceleration (m/s^2) as follows:

$$\Sigma F = ma \quad (1)$$

Next, the torque (τ) on the wheel is expressed as the moment of inertia (kg/m^2) multiplied by the rotational acceleration (m/s^2) in Equation 2.

$$\Sigma \tau = I\alpha \quad (2)$$

The two equations above are closely related in the mathematical equation used to calculate the moment of inertia using Equation 3.

$$I = m_2 R^2 \left(\frac{g}{a_T} - 1 \right) \quad (3)$$

Next, to find the value of the tangential acceleration (a_T), use the following Equation 4:

$$a_T = \frac{2h}{t^2} \quad (4)$$

In the equation above, R represents the radius of the wheel in meters, g represents the acceleration due to gravity of 9.8 m/s^2 , h represents the height of the load in meters, t represents the time taken by the load in seconds, and a_T represents the tangential acceleration in m/s^2 .

METODOLOGY

A. Tools and Materials

The following are the apperatures and materials used in the experiment on the moment of inertia of a rotating wheel:

Table 1. Tools and materials for the rotating wheel moment of inertia experiment.

Apparatus	Quantity	Function
Bicycle wheel and stand	1 set	As the primary system for the experiment
Stopwatch	1 piece	Used to measure the load's travel time
String	1 piece	Used as a medium for transmitting tensile force
Weight	1 set	As the system's drive source
Measuring tape	1 piece	Used to measure the load's height
Digital scale	1 piece	Used to measure the load's mass

B. Work Steps

The following are the procedural steps for the experiment on the moment of inertia of a rotating wheel

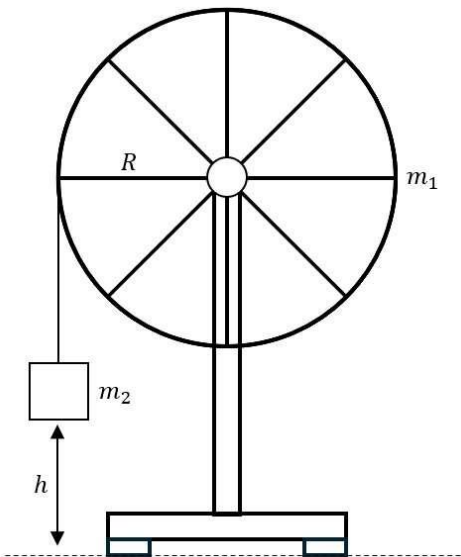


Figure 1. Experimental setup for the moment of inertia of a rotating wheel.

The measurement procedure using a vernier caliper is as follows:

1. Position the bicycle wheel as shown in Figure 1.
2. Measure and record the wheel radius (R).
3. Suspend a weight from the wheel using a string.
4. Determine the height between the weight and the floor, then release the weight.
5. Record the time it takes for the weight to travel the distance h .
6. Perform five repetitions.
7. Repeat the procedure using different weight masses.

D. Flowchart

DATA ANALYSIS AND DISCUSSION

A. Analysis of Measurement Data

The following is a table of data from measurements conducted in the laboratory:

Table 1. Measurement results for a mass variation of 0.05 kg

Repitition	R (m)	h (m)	m (kg)	t (s)
1				
2				
3				
4				
5				

Table 2. Measurement results for a mass variation of 0.1 kg

Repitition	R (m)	h (m)	m (kg)	t (s)
1				
2				
3				
4				
5				

Table 3. Measurement results for a mass variation of 0.15 kg

Repitition	R (m)	h (m)	m (kg)	t (s)
1				
2				
3				
4				
5				

Approved by:

Laboratory Assistant

B. Analysis of Calculation Results Data

Example Calculation:

(NOTE: Find the moment of inertia of the rotating wheel)

The following is a table of the calculation results obtained:

Table 4. Calculation results for a mass variation of 0.05 kg

Repetition	R (m)	h (m)	m (kg)	t (s)	a (m/s ²)	I _p (kg/m ²)
1						
2						
3						
4						
5						
Average						
Standard Deviation						
Uncertainty						

Table 5. Calculation results for a mass variation of 0.1 kg

Repetition	R(m)	h (m)	m (kg)	t (s)	a (m/s ²)	I _p (kg/m ²)
1						
2						
3						
4						
5						
Average						
Standard Deviation						
Uncertainty						

Table 6. Calculation results for a mass variation of 0.15 kg

Repetition	R(m)	h (m)	m (kg)	t (s)	a (m/s ²)	I _p (kg/m ²)
1						
2						
3						
4						
5						
Average						
Standard Deviation						
Uncertainty						

C. Discussion

CONCLUSION

From this experiment, it can be concluded that

- 1.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, Fundamentals of Physics, 10th ed. Hoboken, NJ: John Wiley & Sons, 2014.
- [2] R. A. Serway and J. W. Jewett, Physics for Scientists and Engineers with Modern Physics, 9th ed. Boston, MA: Brooks/Cole Cengage Learning, 2014.
- [3] LD Didactic GmbH, "Definition of moment of inertia," Physics Leaflets P1.4.5.1, Huerth, Germany.
- [4] LD Didactic GmbH, "Moment of inertia and body shape," Physics Leaflets P1.4.5.2, Huerth, Germany.

PRELIMINARY ASSIGNMENT

1. What is meant by the moment of inertia, and how does it affect the rotational motion of an object?
2. What is the principle of Newton's Second Law for rotational motion, and how is it applied in the experiment on the moment of inertia of a rotating wheel?
3. Explain how to determine the value of tangential acceleration in this experiment.
4. Explain the factors that influence an object's moment of inertia.
Derive Equation 3.